INTRODUCTION TO INFORMATION GEOMETRY

Surprisal and Shannon Entropy

Week 5 · Lecture 1

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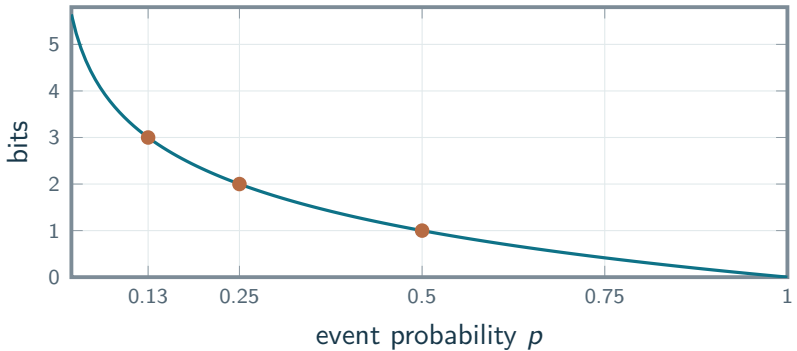
The information in one outcome

An outcome with probability p has surprisal

$$I(p) = -\log_2 p \quad \text{bits.}$$

With natural logarithms, $-\ln p$ measures surprisal in nats. An outcome assigned a smaller probability carries greater surprisal when it occurs.

Surprisal changes nonlinearly



Halving a probability increases surprisal by exactly one bit.

Computing surprisal

An event has probability $p = \frac{1}{8}$. Find its surprisal in bits and in nats:

$$I_2 = -\log_2(\tfrac{1}{8}) = ?, \qquad I_e = -\ln(\tfrac{1}{8}) = ?.$$

Computing surprisal

An event has probability $p = \frac{1}{8}$. Its surprisal in bits and in nats is

$$I_2 = 3 \text{ bits}, \qquad I_e = \ln 8 \text{ nats}.$$

Entropy is average surprisal

For a categorical law $P = (p_1, \ldots, p_n)$,

$$H(P) = -\sum_i p_i \log_2 p_i \quad \text{bits.}$$

This is the expected information of one random draw. The convention $0 \log 0 = 0$ makes entropy continuous at the boundary of a finite simplex.

Entropy of a three-outcome law

Let $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Evaluate

$$H(P) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{4}\log_2\frac{1}{4} - \frac{1}{4}\log_2\frac{1}{4}.$$

Find $H(P)$ in bits.

Entropy of a three-outcome law

Let $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. The three terms are $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ bits.

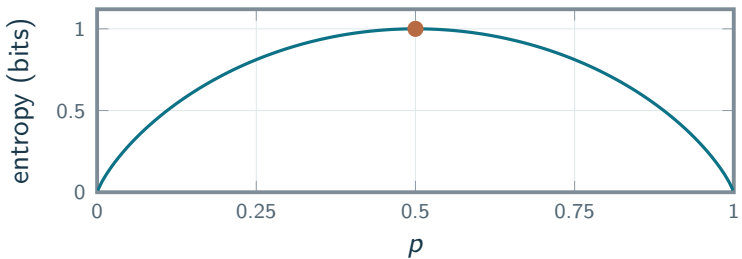
$$H(P) = \frac{3}{2} \text{ bits.}$$

The uniform three-outcome law has a larger entropy, $\log_2 3$.

Binary entropy

For $P_p = (p, 1-p)$,

$$h(p) = -p \log_2 p - (1-p) \log_2 (1-p).$$



Where is binary entropy greatest?

Using natural logs, let

$$h(p) = -p \ln p - (1-p)\ln(1-p).$$

Compute $h'(p)$, solve $h'(p) = 0$, and check concavity.

$$h'(p) = ?, \qquad h''(p) = ?.$$

Where is binary entropy greatest?

Using natural logs, let

$$h(p) = -p \ln p - (1-p)\ln(1-p).$$

Then

$$h'(p) = \ln \frac{1-p}{p}, \qquad h''(p) = -\frac{1}{p(1-p)} < 0.$$

The unique maximum occurs at $p = \frac{1}{2}$.

Uniform distributions maximize entropy

For n possible outcomes,

$$H(P) \leq \log_2 n.$$

Equality holds for $p_i = 1/n$ for all i. The maximum depends on how many outcomes the model allows.

Entropy changes along a trajectory

Let $P(t)$ be a differentiable path of interior distributions. Using natural logarithms,

$$H(t) = -\sum_i p_i(t) \ln p_i(t).$$

Because $\sum_i p_i' = 0$,

$$\frac{dH}{dt} = -\sum_i p_i' \ln p_i.$$

The entropy rate at one state

For a Bernoulli path $p(t)$,

$$\frac{dH}{dt} = \ln \frac{1-p}{p}\, p'.$$

At an instant where $p = \frac{1}{4}$ and $p' = 0.12$, find dH/dt.

The entropy rate at one state

For a Bernoulli path $p(t)$,

$$\frac{dH}{dt} = \ln \frac{1-p}{p}\, p'.$$

At $p = \frac{1}{4}$, $p' = 0.12$,

$$\frac{dH}{dt} = 0.12 \ln 3 > 0 \quad \text{nats/time.}$$

Two uses of the word information

Surprisal measures the information associated with one realized outcome. Entropy averages that quantity over all possible outcomes. Both depend on the chosen probability model.